

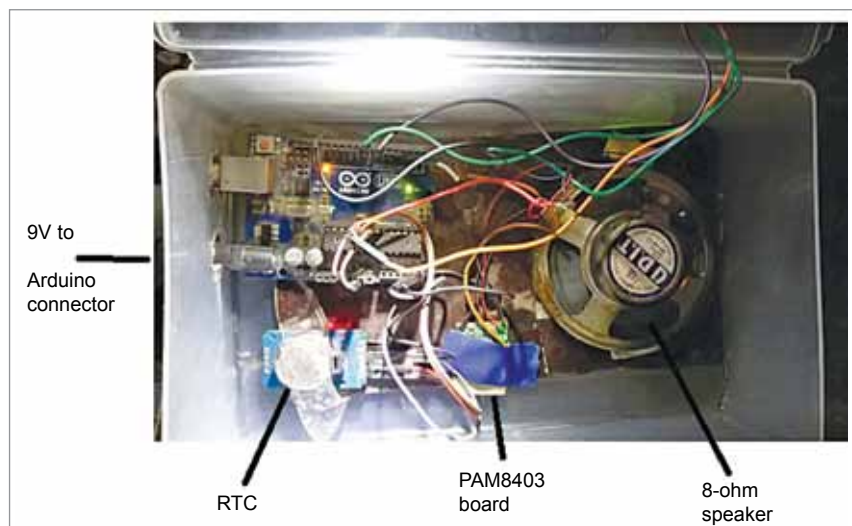


Fig. 4: The photo of the author's unit showing the parts and display

respectively. The RTC has a 3V lithium round button cell battery like the one used in desktop computers. This fits on a round socket at the back of the module as shown in Fig. 4 and works for a year or more. Since the module has a battery, the real time is maintained even when the Arduino is off. Circuit diagram of talking alarm clock is shown in Fig. 3 and prototype setup

is shown in Fig. 4.

The program needs talkie.h library file, which is available at <https://github.com/going-digital/Talkie>

It includes the US English vocabulary. The library file must be included in libraries folder of Arduino IDE as well as the RTCLib library as shown in Fig. 5.

Go to Sketch/Include Library and search for the RTCLib library to include it in the Arduino IDE.

The RTC module keeps time accurately from the moment it is set for the local computer clock, which is ensured while programming the board. Note that the given Arduino code can be used for DS1307 and DS3231 RTC chips without any change in the code. (EFY Lab checked with DS1307 and it worked well.)

The program is set to speak out time once every minute. For 4.51 AM, it will announce, "The time is four fifty one." Alarm time is set in the program itself and one-minute buzzer

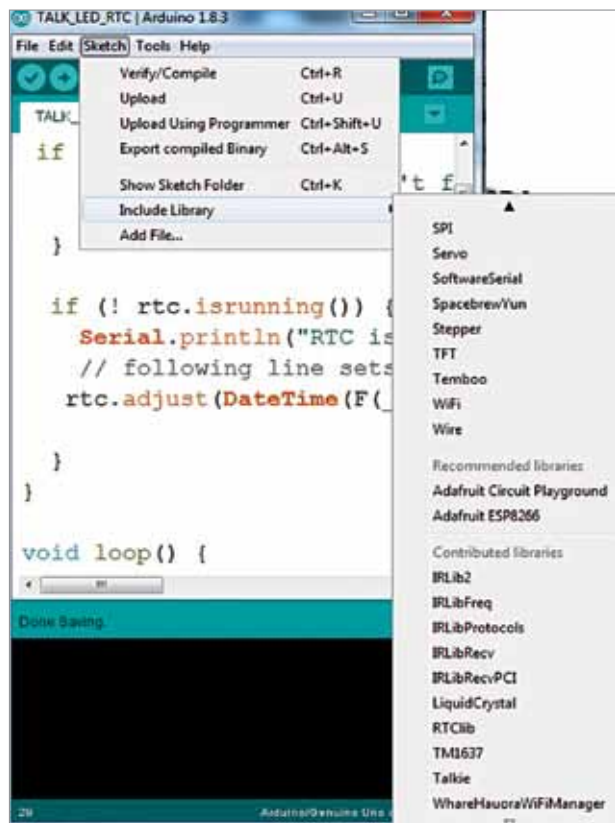


Fig. 5: Including talkie.h file in the library

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- DIS1 - TM1637 LED module
- RTC - DS3231 real-time clock module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm

Miscellaneous:

- LS1 - 8-ohm, 0.5W to 3W loudspeaker
- CON1 - 4-pin male Berg connector
- CON2 - 6-pin male Berg connector
- PZ1 - Piezo buzzer
- Sound board - PAM8403 dual 3W amplifier mini board
- 9V DC adaptor

sound operates from pin 9 of Arduino. The audio program uses Linear Predictive Codec for storing speech bits, which it combines and outputs while telling time in hours and minutes.

The wire from digital pin 3 of Arduino gives sound output as a PWM signal of 10kHz frequency. This is to be filtered and amplified. Filtering takes place automatically in the amplifier. The alarm output is taken from pin 9 of Arduino, to which piezo buzzer PZ1 is connected with a series 220-ohm resistor to limit current.

In order to get a good sound and also minimise on parts size, we recommend the PAM8403 mini audio board (Fig. 7). No external components are needed, just connect pin 3 of Arduino to PAM8403 board and provide 5V supply to the board. Connect the speaker to anyone of the two points marked R (right) and L (left) in the PAM8403 board depending on to which pin the connection from digital pin 3 of Arduino board is given to the input side - R or L. The PAM8403 board is the size of a one-rupee coin and is thin, because all parts on this